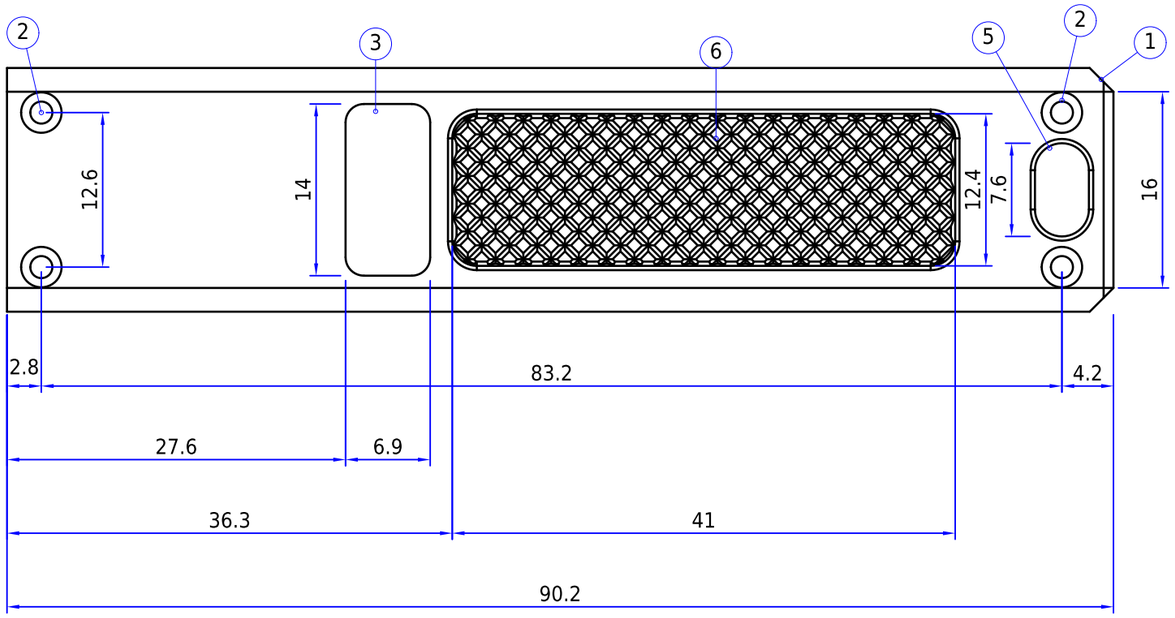
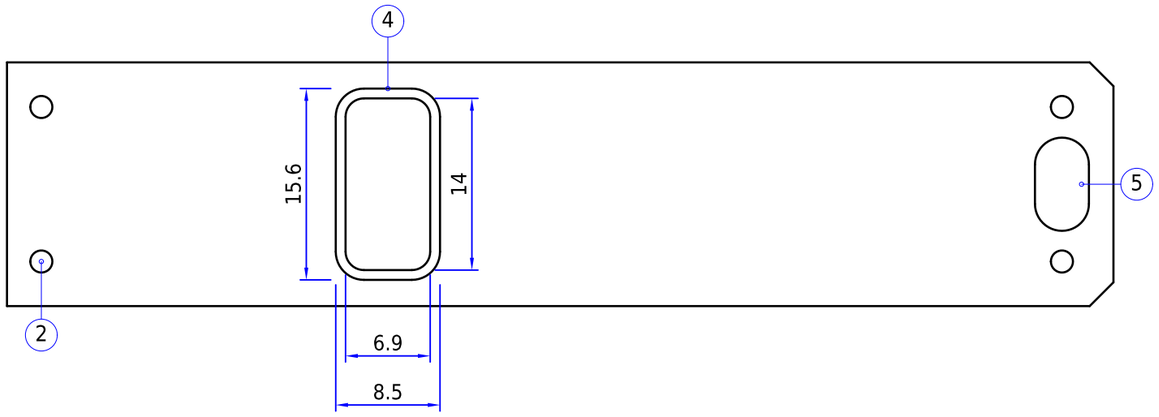
TOP VIEW (from +Z) items 1, 2, 3, 5, 6

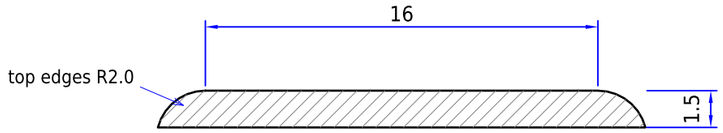


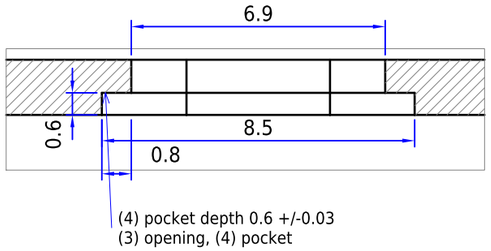
UNDERSIDE (from -Z) items 2, 3, 4, 5



ID	FEATURE	SPECIFICATION	TOLERANCE / NOTE
1	Lid plate	1.5 thick, X 4.2 to 94.4 (body coordinates); top edges R2.0 (flat top 16.0 wide), front edge square; tail facets 2.0 x 45 deg as the tray; underside flat	ISO 2768-mK
2	Screw holes 4x	for M1.6 countersunk (ISO 7046): Ø1.8 thru, countersink Ø3.3 x 90 deg (0.75 deep) on the top face; X 7.0 and 90.2, Y +/- 6.3 (body coordinates)	position +/-0.05 Ø +0.1/0
3	RF window opening	6.9 x 14.0, corners R1.5, through; X 31.8 to 38.7 (body coordinates), centred on Y 0 (over the ESP32 module antenna)	+/-0.05
4	RF pocket (underside)	8.5 x 15.6, corners R2.3, 0.6 deep (Z 14.5 to 15.1), 0.8 margin round the opening; seat of the resin insert flange; 0.9 of lid remains above the pocket	+/-0.05; depth +/-0.03
5	Lanyard slot, THRU	stadium 4.4 x 7.6 (R2.2 ends), centre X 90.2, Y 0; chamfer C0.35 on the top face; same slot as the tray (AUR-H-01 item 19)	+/-0.1
6	Knurl panel (top face)	X 40.5 to 81.5, 12.4 wide, R2.0, depth 0.35: the same pattern as the tray bottom (AUR-H-01 item 17, detail K); mirrored on the top face	depth +/-0.05
7	RF window insert	not machined here: translucent resin part (Aurora_rf_window.step), 0.05 clearance in the opening and the pocket, glued in from inside	
8	Finish	glass-bead blast, then black anodise type II, 8-12 um (as the tray)	

SECTION J-J (X = 20, no knurl) item 1





SECTION G-G (Y = 0, no knurl) items 3, 4

SECTION H-H (Y = 6.3, front screw) item 2

NOTES
1 Material: aluminium 6061-T6, 1.5 thick. Geometry per Aurora_lid.step; X, Y, Z are body coordinates (X = 0 at the front face of the body).
2 General tolerances ISO 2768-mK; toleranced values are critical.
3 Screws: 4x M1.6 x 4 countersunk, ISO 7046 / DIN 965, A2 stainless. The countersink edge is only 0.05 inside the start of the edge radius (Y 8.0): do not enlarge it.
4 Finish: fine glass-bead blast, then anodise type II matte black (8-12 um), as the tray. Cut the knurl before blasting.
5 Sections G-G, H-H, J-J are drawn without the knurl.
6 Thin spots in the STEP (mm): 1.15 lid under the knurl; 0.9 lid above the RF pocket. The knurl panel starts at X 40.5, clear of the pocket (ends at X 39.5): keep it there.
7 At X 90.2 the rear countersinks and the slot chamfer leave a sliver of only 0.5 at the top face: break the edges lightly (max 0.05) and do not deepen the chamfers.
8 The RF window insert (item 7) is a separate printed part; opening and pocket sizes must not change without the insert.

LID

Part: AUR-H-02 Aurora_lid.step	SCALE 2:1 (details as marked) SHEET 1/1
Aurora++ keychain flashlight v0.5 2026-10-14	Units mm | ISO 2768-mK